Band-1 Computational Experiment: No-Three-in-Line — Exact Maxima for Small Grids

Human + ZoraASI full-dash (executor)

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Honesty stamp. This note does **not** claim the asymptotic no-three-in-line problem is solved. The Guy–Kelly heuristic ($D(n)/n \to \pi/\sqrt{3}$) and the Hall–Jackson–Sudbery–Wild constructive lower bound $\sim (3/2)n$ leave a gap below the trivial upper bound $2n$. Band-1 here is a **literature ledger** of known $D(n) = 2n$ existence plus **box-certified exact values** for small n by search (backtrack / SAT). Settling all n, or the asymptotic density, is **not** claimed.

Abstract

Let $D(n)$ be the maximum number of points selectable from the $n \times n$ integer grid with no three collinear. Always $D(n) \leq 2n$. We (i) ledger the 2026 literature status of $D(n) = 2n$ existence; (ii) certify exact $D(n)$ on-box for $n \leq 12$ (with $D(1) = 1$) by finding size-$2n$ configurations (verified) together with the trivial upper bound; (iii) record the **next open grid for $2n$-existence:** $n = 75$. Asymptotic status remains **OPEN**.

Problem statement

Select as many points as possible from $\{0, \ldots, n-1\}^2$ so that no three lie on a Euclidean line. Dudeney (1906/1917) asked for $2n$ on the chessboard; the general problem is classical (Guy–Kelly 1968; Flammenkamp 1992–2026).

Literature ledger (not re-proved on box)

Item	Status (survey 2026-09-11)	Source
Upper bound	$D(n) \leq 2n$ always	pigeonhole (at most 2 per row)
$D(1)$	**1**	trivial
$D(n) = 2n$ for $2 \leq n \leq 74$	known (existence of $2n$-configs)	Flammenkamp DB; Prellberg; classical
$D(76) = 152$	known	Heule SAT (Aug 2026)
Next open $2n$-existence	$n = 75$	MathWorld / Flammenkamp table 2026-08-19
Larger sporadic	$n \geq 77$ not generally settled	same
Best general lower bound	$(3/2 - \varepsilon)n$ for large n	Hall–Jackson–Sudbery–Wild, JCTA 1975

Item	Status (survey 2026-09-11)	Source
Guy–Kelly heuristic	$D(n)/n \to \pi/\sqrt{3} \approx 1.8138$	probabilistic; **not a theorem**
Full enum of solutions	complete for $n \leq 20$ (all symmetries)	Flammenkamp / Chaffin / Riley / Prellberg
Asymptotic / for all n	**OPEN**	—

Recent computational extensions (Prellberg CP-SAT; Heule SAT: $n \in \{65, 67, 69, 70, 71, 72, 73, 76\}$) closed all $2n$-gaps through 74 and added 76; **75 remains the smallest n without a published $2n$ configuration**.

Method (box)

1. **Upper bound.** $D(n) \leq 2n$ (at most two points in any row).
2. **Backtrack.** Place exactly two points per row, at most two per column; incremental line-occupancy pruning; verify no three collinear by exact integer cross-product.
3. **SAT (Glucose3 via PySAT).** Exactly-two-per-row/column cardinality + at-most-two on every grid line with at least 3 points.
4. **Certificate.** A verified size-$2n$ set implies $D(n) = 2n$ for $n \geq 2$.
5. **Stop.** No overnight hang on $n \geq 13$ without a fast construction; literature covers those n through 74.

Results (numbers!)

Unit tests

`PASS`: collinearity detector; small-n search sanity (`results.json`).

Box-certified exact values

n	$D(n)$ (box)	$2n$	Justification
1	**1**	2	trivial cell
2	**4**	4	box BT/SAT + upper bound
3	**6**	6	box BT/SAT + upper bound
4	**8**	8	box BT/SAT + upper bound
5	**10**	10	box BT/SAT + upper bound
6	**12**	12	box BT/SAT + upper bound
7	**14**	14	box BT/SAT + upper bound
8	**16**	16	box BT/SAT + upper bound
9	**18**	18	box BT/SAT + upper bound
10	**20**	20	box BT/SAT + upper bound
11	**22**	22	box BT/SAT + upper bound
12	**24**	24	box BT/SAT + upper bound

Certified tonight: $D(1) = 1$ and $D(n) = 2n$ for all $n = 2, \ldots, 12$.

Not pushed further on-box tonight: $n \in \{13, \ldots, 16\}$ (tractable with more time / known constructions, but not required for the ledger claim).

Sample certificates

Verified configurations for $n \leq 12$ are stored in `configurations.json`. Each stored set was re-checked: no three collinear; size $= 2n$.

Literature next open

Next open grid size for $D(n) = 2n$ existence: $n = 75$ (want 150 points). Largest listed $2n$ configuration in the surveyed sources: $n = 76$ (Heule).

Why Band-1 vs asymptotic

- **Band-1 (this note):** exact finite-n maxima + ledger of which n have $2n$-solutions; incremental computational frontier.
- **Not claimed:** $D(n) = 2n$ for all n; Guy–Kelly constant; closing the $(3/2)n$–$2n$ asymptotic gap.

Reproducibility

- Code: `/workspace/experiments/band1/no_three_in_line/`
- Run: `python3 ntil_exact.py`
- Artifacts: `results.json`, `literature_ledger.json`, `configurations.json`, `SUMMARY.txt`
- Log: `/workspace/experiments/logs/2026-09-11_no_three_in_line.log`

Conclusion

Box: exact $D(1) = 1$, $D(n) = 2n$ for $2 \leq n \leq 12$, via search certificates + trivial upper bound.

Literature: $D(n) = 2n$ known for all $2 \leq n \leq 74$ and $n = 76$; **next open $2n$-existence grid** $n = 75$.

Asymptotic problem: OPEN — not claimed solved.